

Lent Week 7 & 8

lw703

1 Completed Task

We derive the MALA transition kernel, $T(\mathbf{y}|\mathbf{x})$:

$$T(\mathbf{y} | \mathbf{x}) = q(\mathbf{y} | \mathbf{x}) \alpha(\mathbf{x}, \mathbf{y}) + \delta_{\mathbf{x}}(\mathbf{y}) \left(1 - \int q(\mathbf{y}' | \mathbf{x}) \alpha(\mathbf{x}, \mathbf{y}') d\mathbf{y}' \right). \quad (1)$$

Here, $q(\mathbf{y} | \mathbf{x})$ is the proposal distribution density for suggesting state $\mathbf{y}$ from state $\mathbf{x}$, $\alpha(\mathbf{x}, \mathbf{y})$ is the Metropolis-Hastings acceptance probability, and $\delta_{\mathbf{x}}(\mathbf{y})$ is the Dirac delta function, which accounts for the probability of the chain remaining at state $\mathbf{x}$ if the proposed move $\mathbf{y}'$ is rejected.

The proposal distribution $q(\mathbf{y} | \mathbf{x})$ is generated using a Langevin sampler:

$$q(\mathbf{y} | \mathbf{x}) = p_{\gamma}(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}; \mathbf{x} + \gamma(\mathbf{x})\nabla \log p_d(\mathbf{x}), 2\gamma(\mathbf{x})\mathbf{I}). \quad (2)$$

In this formulation, $\mathbf{x}$ is the current sample, $\nabla \log p_d(\mathbf{x})$ is the score function of the target data distribution $p_d(\mathbf{x})$, and $\mathbf{I}$ is the identity matrix, signifying isotropic noise for the proposal.

The acceptance probability $\alpha(\mathbf{x}, \mathbf{y})$ is determined by the Metropolis-Hastings ratio, $r(\mathbf{x}, \mathbf{y})$:

$$r(\mathbf{x}, \mathbf{y}) = \frac{p_d(\mathbf{y}) q(\mathbf{x} | \mathbf{y})}{p_d(\mathbf{x}) q(\mathbf{y} | \mathbf{x})}, \quad (3)$$

where $q(\mathbf{x} | \mathbf{y})$ is the probability density of proposing to move from $\mathbf{y}$ back to $\mathbf{x}$, calculated analogously to Equation 2. The acceptance probability is then:

$$\alpha(\mathbf{x}, \mathbf{y}) = \min\{1, r(\mathbf{x}, \mathbf{y})\}. \quad (4)$$

This mechanism ensures detailed balance with respect to $p_d(\mathbf{x})$, guaranteeing convergence to the target distribution.

To resulting marginal:

$$p_{\gamma|\theta}^m(\mathbf{y}) = \int T_{\gamma}(\mathbf{y}|\mathbf{x}) p_{\theta}(\mathbf{x}) d\mathbf{x}. \quad (5)$$

The optimization problem for $\gamma(\mathbf{x})$ is thus formulated as:

$$\gamma^*(\cdot) = \operatorname{argmin}_{\gamma(\cdot)} \mathbb{D}_{\text{KL}}(p_{\gamma|\theta}^m \| p_d). \quad (6)$$

Same as Langevin sampler:

$$\nabla_{\gamma} \mathbb{D}_{\text{KL}}(p_{\gamma|\theta}^m \| p_d) = E_{\mathbf{x} \sim p_{\theta}(\mathbf{x})} E_{\mathbf{y} \sim T_{\gamma}(\cdot|\mathbf{x})} \left[\left(\nabla_{\mathbf{y}} \log p_{\gamma|\theta}^m(\mathbf{y}) - \nabla_{\mathbf{y}} \log p_d(\mathbf{y}) \right) \frac{\partial \mathbf{y}}{\partial \gamma} \right]. \quad (7)$$

Analyzing the score of the continuous part of the MALA kernel (i.e., for accepted moves, $\mathbf{y} \neq \mathbf{x}$), we consider $\nabla_{\mathbf{y}} \log (q(\mathbf{y} | \mathbf{x}) \alpha(\mathbf{x}, \mathbf{y}))$. This can be decomposed as:

$$\nabla_{\mathbf{y}} \log (q(\mathbf{y} | \mathbf{x}) \alpha(\mathbf{x}, \mathbf{y})) = \nabla_{\mathbf{y}} \log q(\mathbf{y} | \mathbf{x}) + \nabla_{\mathbf{y}} \log \alpha(\mathbf{x}, \mathbf{y}). \quad (8)$$

The gradient of the log acceptance probability, $\nabla_{\mathbf{y}} \log \alpha(\mathbf{x}, \mathbf{y})$:

$$\nabla_{\mathbf{y}} \log \alpha(\mathbf{x}, \mathbf{y}) = \begin{cases} \nabla_{\mathbf{y}} \log r(\mathbf{x}, \mathbf{y}) & \text{if } r(\mathbf{x}, \mathbf{y}) < 1, \\ 0 & \text{if } r(\mathbf{x}, \mathbf{y}) > 1, \\ \text{undefined} & \text{if } r(\mathbf{x}, \mathbf{y}) = 1. \end{cases} \quad (9)$$

Substituting $\log r(\mathbf{x}, \mathbf{y}) = \log p_d(\mathbf{y}) + \log q(\mathbf{x} \mid \mathbf{y}) - \log p_d(\mathbf{x}) - \log q(\mathbf{y} \mid \mathbf{x})$, the score of an accepted move (continuous part of the kernel) becomes:

$$\nabla_{\mathbf{y}} \log (q(\mathbf{y} \mid \mathbf{x}) \alpha(\mathbf{x}, \mathbf{y})) = \begin{cases} \nabla_{\mathbf{y}} \log p_d(\mathbf{y}) + \nabla_{\mathbf{y}} \log q(\mathbf{x} \mid \mathbf{y}) & \text{if } r(\mathbf{x}, \mathbf{y}) < 1, \\ \nabla_{\mathbf{y}} \log q(\mathbf{y} \mid \mathbf{x}) & \text{if } r(\mathbf{x}, \mathbf{y}) > 1, \\ \text{undefined} & \text{if } r(\mathbf{x}, \mathbf{y}) = 1. \end{cases} \quad (10)$$